

Econ 2 - Lecture 10 - 2/11/26

Discussion Activity #3 This Week!

Lecture Quiz #5 Posted Today, Due Wednesday, 2/18

No Class Monday, 2/16

Last Classes:

→ Defined Equilibrium as $Y = AE$ (Output = Spending)

→ Consumption is the largest component of AE

→ Income (Y) is the biggest determinant of C

→ $C = AC + MPC \cdot (Y - T)$

Real GDP (Y)	Taxes (T)	Disposable Income ($Y - T$)	Consumption (C)
200	100	100	180
300	100	200	260
400	100	300	340
500	100	400	420
600	100	500	500
700	100	600	580
800	100	700	660
900	100	800	740

What is MPC ? What is AC ?

$$C = 100 + 0.8(Y - T)$$

$$MPC = \frac{\Delta C}{\Delta Y - T} \Rightarrow \frac{\Delta C}{\Delta Y} = \frac{580 - 500}{700 - 600} = \frac{80}{100} = 0.80$$

$$AC? \quad C = AC + MPC(Y - T)$$

$$180 = AC + 0.8 \cdot (200 - 100)$$

$$180 = AC + 0.8(100) = AC + 80 \rightarrow \boxed{AC = 100}$$

What about other groups of spenders?

Next Group = Firms

1. Capital/Machines
2. New Homes } Planned Investment Spending (I^P)

Not include unplanned changes in inventory
→ Not considered spending

I^P is not always equal to original I (from) GDP

Assume I^P is fixed at all levels of Y
↳ same as T & AC

Next Group of Spenders: Government

Government Spending = G

Not including wealth transfers, unemployment benefits
↳ no production, not spending (AE),
not $GDP (Y)$

Include goods/services that are necessary but
hard to profit from: National Defence, Police,
Fire, Infrastructure, Education,
etc.

Assume G is fixed for all levels of Y

Last Group: Foreigners

$$\text{Net Exports (NX)} = \text{Exports (X)} - \text{Imports (M)}$$

Possible for $NX < 0$, $M > X$

Build the Aggregate Expenditure Equation

$$AE = C + I^P + G + NX$$

What does mean for $Y = AE$?

$$\underbrace{\cancel{C} + I + \cancel{G} + \cancel{NX}}_Y = \underbrace{\cancel{C} + I^P + \cancel{G} + \cancel{NX}}_{AE}$$

$$I = I^P$$

~~1. Capital~~

~~2. New Homes~~

3. Δ in
Inventory

~~1. Capital~~

~~2. New Homes~~

3. $\emptyset$

If $Y = AE$, change in inventory = $\emptyset$

If change in inventory = $\emptyset$

→ No extra production

→ No need to go to "stockroom" to meet demand

→ Sell exactly the amount they produce

⇒ Equilibrium

Show $Y = AE$ graphically

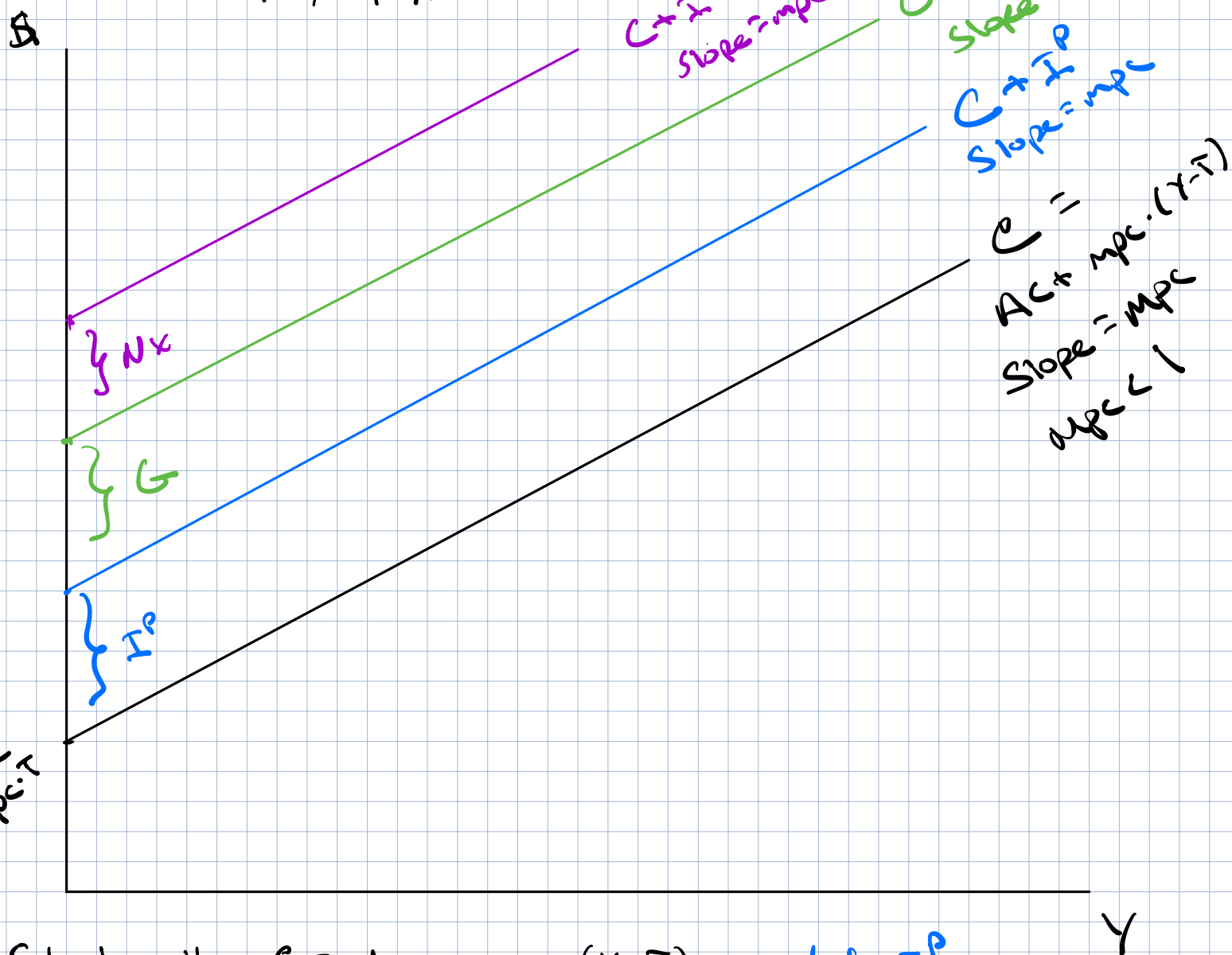
Start by graphing AE

$$AE = C + I^P + G + NX$$

$$= AC + mpc \cdot (Y - T) + I^P + G + NX$$

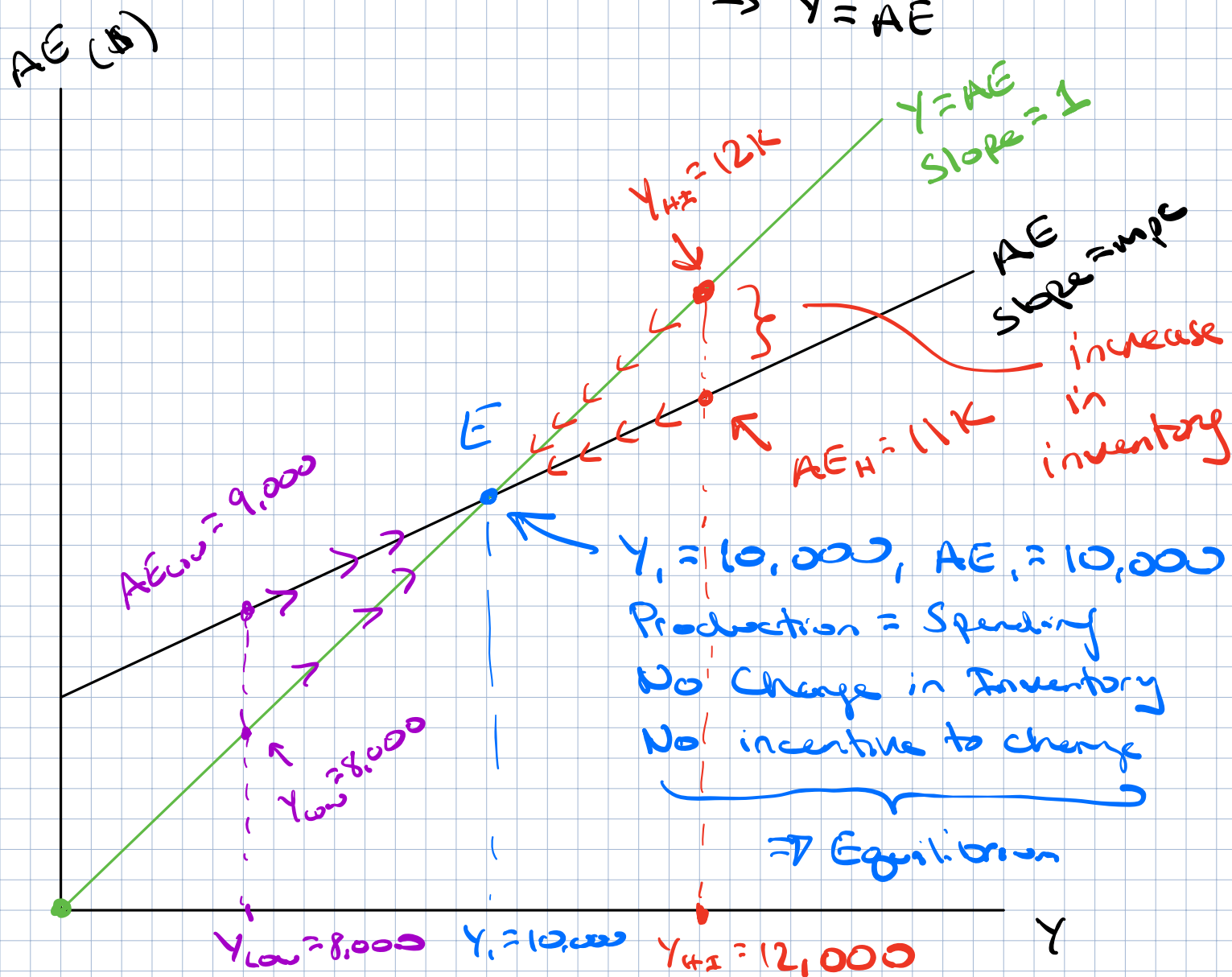
$\downarrow$ Fixed $\downarrow$ Constant $\downarrow$ Fixed $\downarrow$ Fixed $\downarrow$ Fixed $\downarrow$ Fixed

$$AE = \underbrace{\text{Fixed Values}}_{AC, T, I^P, G, NX} + mpc \cdot Y$$



Start with $C = AC + mpc \cdot (Y - T)$, add I^P ,
add G , add NX

Drawn out AE $\rightarrow$ need to show equilibrium
 $\rightarrow Y = AE$



Add curve $\rightarrow Y = AE$ (Equilibrium - Line)

If $Y = 0, AE = 0 \rightarrow$ origin, slope = 1

If $Y = 100, AE = 100, Y = 12345, AE = 12345$

Start at $Y_{Hx} = 12,000, AE_{Hx} = 11,000$

Production > Spending, $Y > AE \rightarrow$ Too much Y

Increase in inventory $\rightarrow 12K - 11K = 1,000$

Change in inventory = $Y - AE$

Incentive for firms to decrease $Y \rightarrow$ not in equilibrium

Start at $Y_{cur} = 8,000$, $AE_{cur} = 9,000$

Spending > Production, $AE > Y$, Not enough Y

Decrease in inventory $\rightarrow$ Change in inventory

$$= Y - AE = 8000 - 9000 = -1000$$

Incentive for firms to increase $Y \rightarrow$ not in equilibrium

Find $Y = AE \rightarrow$ Table

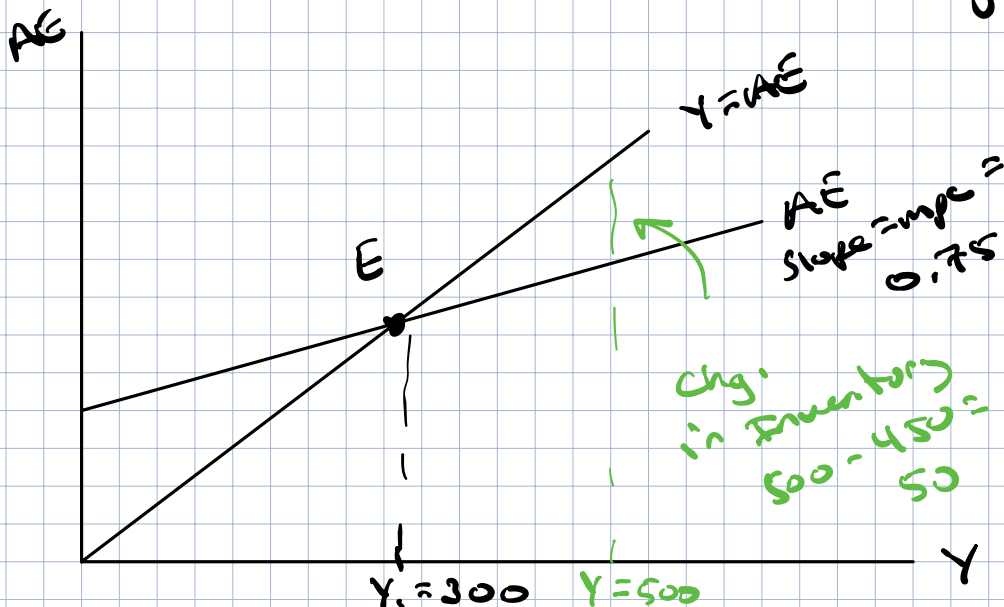
Y	$C + 25$	I^p	G	NX	AE	$Y - AE$	Chg. in Inventory
200	200	10	10	5	225	-25	-25
300	275	10	10	5	300	0	0
400	350	10	10	5	375	+25	+25
500	425	10	10	5	450	+50	+50
600	500	10	10	5	525	+75	+75

What is equilibrium level of Y ?

$$AE = C + 10 + 10 + 5 = C + 25$$

$\downarrow$ $\downarrow$ $\downarrow$
 I^p G NX

When $Y = 300 \rightarrow AE = 300 \rightarrow$ Equilibrium



Learned
 Δ in Inventory
 $= 50$,
What is Y ?

$$\text{Chg. in Inventory} = 500 - 450 = 50$$

$T = 100$, Value of AC ?

$$C = AC + mpc \cdot (Y - T)$$



$$200 = AC + 0.75(200 - 100)$$

$$200 = AC + 0.75(100)$$

$$200 = AC + 75$$

$$AC = 125$$